

Polynomial Interpolation (piecewise)

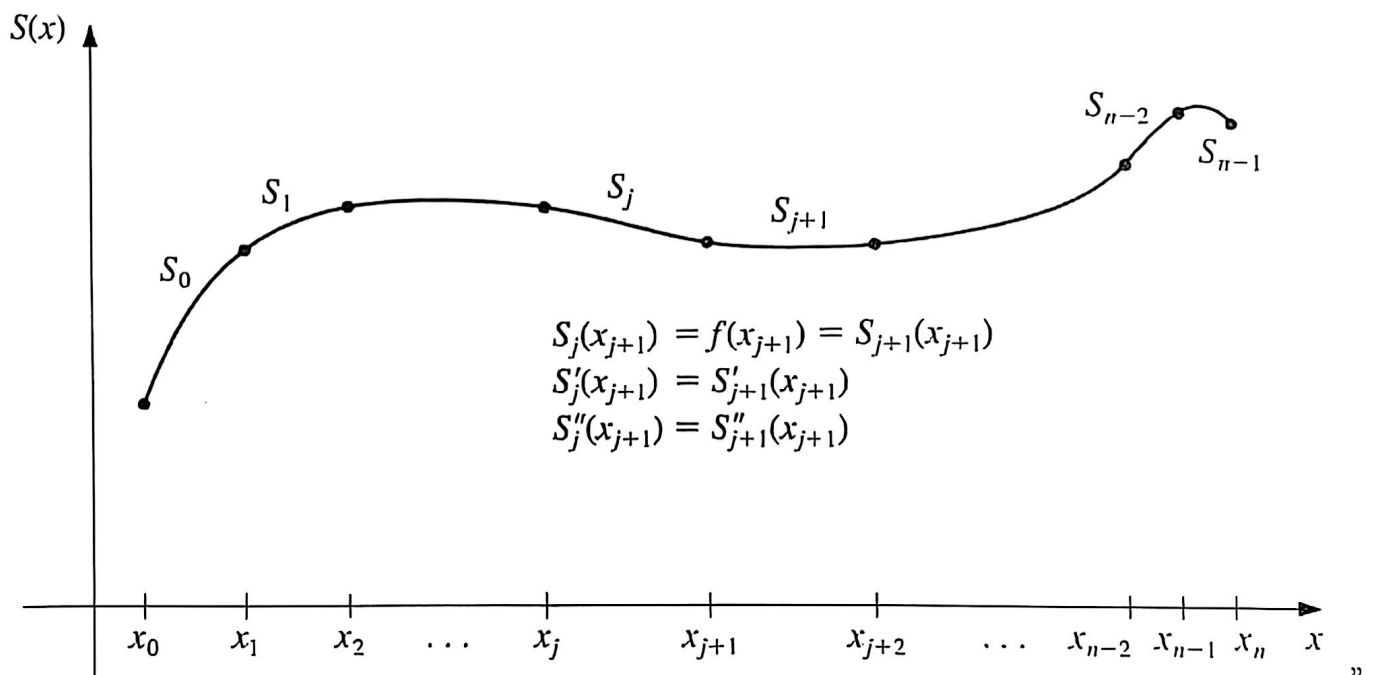
1.3 Cubic Splines

Interpolation relying on a single high-degree polynomial—such as in Lagrange interpolation—can lead to erratic oscillations, especially when the number of points exceeds a dozen. This phenomenon highlights why very high-degree polynomials should generally be avoided in favor of piecewise or spline interpolation for better stability and accuracy.

In piecewise interpolation, we divide the interval into small subintervals and construct a low degree (0 to 3) interpolation polynomial on each subinterval, which is called a **spline**.

1. **Piecewise Linear Interpolation:** Uses straight lines to connect points but is not differentiable at the interpolating points.
2. **Piecewise Quadratic Interpolation:** Uses quadratic functions on each subinterval. However, enforcing derivative conditions at boundary points x_0 and x_n can be problematic.
3. **Cubic Spline:** A degree-3 spline that is twice differentiable at interpolation points, with well-defined boundary conditions. It is the most widely used piecewise polynomial approximation due to its smoothness and stability.

A simple Definition of Cubic Spline Interpolation: Given a set of data points, a cubic spline interpolant is a piecewise cubic function that passes through the given points while maintaining continuous first and second derivatives (C^2 continuity) across the entire domain.



Detailed Definition: Cubic Splines

Given as set of data (x_i, y_i) , with $a = x_0 < x_1 < \dots < x_n = b$,

- (a) $S(x)$ is a cubic polynomial, denoted by $S_j(x)$, on the subinterval $[x_j, x_{j+1}]$ for each $j = 0, \dots, n-1$.

- (b) $S_j(x_j) = f(x_j)$ and $S_j(x_{j+1}) = f(x_{j+1})$, $j = 0, \dots, n-1$.

the curve should go through the given data points

- (c) $S_j(x_{j+1}) = S_{j+1}(x_{j+1})$, $j = 0, \dots, n-2$.

the two cubic polynomials S_j & S_{j+1} should be the same at x_{j+1}



- (d) $S'_j(x_{j+1}) = S'_{j+1}(x_{j+1})$, $j = 0, \dots, n-2$.

the curve could be continuous but not smooth
to avoid this, we need the 1st derivatives to exist and be the same at x_{j+1}



- (e) $S''_j(x_{j+1}) = S''_{j+1}(x_{j+1})$, $j = 0, \dots, n-2$.

We also need to remove any sudden changes in the curvature

- (f) One of the following sets of boundary conditions (BCs) is imposed:

(i) $S''(x_0) = S''(x_n) = 0$, natural or free BC

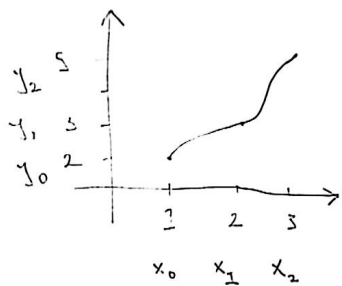
(ii) $S'(x_0)$ and $S'(x_n)$ are equal to some values; clamped BC.

} what will happen at the end points

When the natural BC is used, the result is **natural cubic spline**, and its graph approximates the shape, which a long flexible rod would assume if forced to go through each of the data points. Similarly, when the clamped boundary conditions occur, the spline is called a **clamped cubic spline**. In general, clamped boundary conditions lead to more accurate approximations since they include more information about the function.

To construct the cubic line interpolant for a given function f , the conditions in the above definition are applied to the cubic polynomials. There is a unique natural cubic spline for any data set. That is, the cubic spline S interpolating (x_i, y_i) with $a = x_0 < \dots < x_n = b$ and satisfying $S''(a) = S''(b) = 0$.

Example 1. Construct a natural cubic spline passing through (1, 2), (2, 3), and (3, 5).



$$\text{on } [1, 2]: S_0(x) = a_0 + b_0(x-x_0) + c_0(x-x_0)^2 + d_0(x-x_0)^3$$

$$\text{on } [2, 3]: S_1(x) = a_1 + b_1(x-x_1) + c_1(x-x_1)^2 + d_1(x-x_1)^3$$

① must agree at the data points (b)

$$S_0(x_0) = y_0 \Rightarrow S_0(1) = 2 \Rightarrow a_0 = 2$$

$$S_0(x_1) = y_1 \Rightarrow S_0(2) = 3 \Rightarrow a_0 + b_0 + c_0 + d_0 = 3$$

$$S_1(x_1) = y_1 \Rightarrow S_1(2) = 3 \Rightarrow a_1 = 3$$

$$S_1(x_2) = y_2 \Rightarrow S_1(3) = 5 \Rightarrow a_1 + b_1 + c_1 + d_1 = 5$$

② first derivative must agree (c)

$$S'_0(x) = b_0 + 2c_0(x-1) + 3d_0(x-1)^2$$

$$S'_1(x) = b_1 + 2c_1(x-2) + 3d_1(x-2)^2$$

$$S'_0(x_1) = S'_1(x_1)$$

$$S'_0(2) = S'_1(2)$$

$$b_0 + 2c_0 + 3d_0 = b_1$$

③ second derivative must agree (e)

$$S''_0(x) = 2c_0 + 6d_0(x-1)$$

$$S''_1(x) = 2c_1 + 6d_1(x-2)$$

$$S''_0(x_1) = S''_1(x_1)$$

$$S''_0(2) = S''_1(2)$$

$$2c_0 + 6d_0 = 2c_1$$

$$S(x) = \begin{cases} S_0(x) & x \in [1, 2] \\ S_1(x) & x \in [2, 3] \end{cases}$$

④ Using natural B.C

$$S''_0(x_0) = 0 \Rightarrow S''_0(1) = 0 \Rightarrow 2c_0 = 0$$

$$S''_1(x_2) = 0 \Rightarrow S''_1(3) = 0 \Rightarrow 2c_1 + 6d_1 = 0$$

$$b_j(a)$$

$$d_{j+1}(x-x_{j+1})$$

$$S_j(x) = a_j + b_j(x-x_j) + c_j(x-x_j)^2 + d_j(x-x_j)^3 \rightarrow S_{j+1}(x) = a_{j+1} + b_{j+1}(x-x_{j+1}) + c_{j+1}(x-x_{j+1})^2 + d_{j+1}(x-x_{j+1})^3$$

$$S'_j(x) = b_j + 2c_j(x-x_j) + 3d_j(x-x_j)^2 \rightarrow S'_{j+1}(x) = b_{j+1} + 2c_{j+1}(x-x_{j+1}) + 3d_{j+1}(x-x_{j+1})^2$$

$$S''_j(x) = 2c_j + 6d_j(x-x_j) \rightarrow S''_{j+1}(x) = 2c_{j+1} + 6d_{j+1}(x-x_{j+1})$$

$$\textcircled{C} S''_j(x_{j+1}) = S''_{j+1}(x_{j+1})$$

$$2c_j + 6d_j(x_{j+1}-x_j) = 2c_{j+1} + 6d_{j+1}(x_{j+1}-x_{j+1})$$

$$6d_j = \frac{2c_{j+1} - 2c_j}{x_{j+1} - x_j}$$

$$= \frac{c_{j+1} - c_j}{3h_j}$$

$$\textcircled{D} S'_j(x_{j+1}) = S'_{j+1}(x_{j+1})$$

$$b_j + 2c_j(x_{j+1}-x_j) + 3d_j(x_{j+1}-x_j)^2 = b_{j+1} + 2c_{j+1}(x_{j+1}-x_{j+1}) + 3d_{j+1}(x_{j+1}-x_{j+1})^2$$

$$b_j + 2c_j h_j + 3d_j h_j^2 = b_{j+1}$$

$$b_j + 2c_j h_j + 3\left(\frac{c_{j+1}-c_j}{3h_j}\right) h_j^2 = b_{j+1}$$

$$b_j + 2c_j h_j + (c_{j+1}-c_j) h_j = b_{j+1}$$

$$b_j + (c_{j+1}+c_j) h_j = b_{j+1} \quad \textcircled{A}$$

$$\textcircled{E} S_j(x_{j+1}) = S_{j+1}(x_{j+1})$$

$$a_j + b_j(x_{j+1}-x_j) + c_j(x_{j+1}-x_j)^2 + d_j(x_{j+1}-x_j)^3 = a_{j+1} + 0$$

$$a_j + b_j h_j + c_j h_j^2 + d_j h_j^3 = a_{j+1}$$

$$a_j + b_j h_j + c_j h_j^2 + \frac{c_{j+1}-c_j}{3h_j} \cdot h_j^3 = a_{j+1}$$

$$a_j + b_j h_j + c_j h_j^2 + \left(\frac{c_{j+1}-c_j}{3}\right) h_j^2 = a_{j+1}$$

$$b_j h_j = a_{j+1} - \left(\frac{c_{j+1}-c_j}{3}\right) h_j^2 - a_j - c_j h_j^2$$

$$b_j h_j = a_{j+1} - a_j - h_j^2 \left(\frac{c_{j+1}-c_j}{3} + c_j\right)$$

$$b_j = \frac{a_{j+1}-a_j}{h_j} - h_j \left(\frac{c_{j+1}-c_j+3c_j}{3}\right)$$

$$= \frac{a_{j+1}-a_j}{h_j} - h_j \left(\frac{2c_j+c_{j+1}}{3}\right)$$

$$b_{j-1} = \frac{(a_j-a_{j-1})}{h_{j-1}} - \frac{h_{j-1}}{3} (2c_{j-1}+c_j)$$

drop (A) by an index

$$b_{j-1} + (c_j+c_{j-1}) h_{j-1} = b_j$$

$$\frac{(a_j-a_{j-1})}{h_{j-1}} - \frac{h_{j-1}}{3} (2c_{j-1}+c_j) + (c_j+c_{j-1}) h_{j-1} =$$

$$\frac{(a_{j+1}-a_j)}{h_j} - \frac{h_j}{3} (2c_j+c_{j+1}) \quad \text{[Multiply through by 3]}$$

$$P. \text{ of cont.} \quad -h_{j-1}(2c_{j-1} + c_j) + 3(c_j + c_{j-1})h_{j-1} + h_j(2c_j + c_{j+1}) = \\ 3\left(\frac{a_{j+1} - a_j}{h_j}\right) - 3\left(\frac{a_j - a_{j-1}}{h_{j-1}}\right)$$

$$c_{j-1}(-2h_{j-1} + 3h_{j-1}) + c_j(-h_{j-1} + 3h_{j-1} + 2h_j) + c_{j+1}h_j = \\ 3\left(\frac{a_{j+1} - a_j}{h_j}\right) - 3\left(\frac{a_j - a_{j-1}}{h_{j-1}}\right)$$

$$h_{j-1}c_{j-1} + 2(h_{j-1} + h_j)c_j + h_jc_{j+1} = \frac{3}{h_j}(a_{j+1} - a_j) - \frac{3}{h_{j-1}}(a_j - a_{j-1})$$

$$\text{when } j=0 \quad S_0''(x) = 2c_0 + 6d_0(x-x_0) \\ \text{by } B(c_i) \quad S_0''(x_0) = 0 \\ \Rightarrow 2c_0 = 0 \\ \Rightarrow c_0 = 0$$

$$\text{when } j=n \quad S_n''(x) = 2c_n + 6d_n(x-x_n) \\ \text{by } B(c_i) \quad S_n''(x_n) = 0 \\ \Rightarrow 2c_n = 0 \\ \Rightarrow c_n = 0$$

$$\text{when } j=1 \quad h_0c_0 + 2(h_0 + h_1)c_1 + h_1c_2 = \frac{3}{h_1}(a_2 - a_1) - \frac{3}{h_0}(a_1 - a_0)$$

$$\text{when } j=2 \quad h_1c_1 + 2(h_1 + h_2)c_2 + h_2c_3 = \frac{3}{h_2}(a_3 - a_2) - \frac{3}{h_1}(a_2 - a_1)$$

$$\text{when } j=n-1 \quad h_{n-2}c_{n-2} + 2(h_{n-2} + h_{n-1})c_{n-1} + h_{n-1}c_n = \frac{3}{h_{n-1}}(a_n - a_{n-1}) - \frac{3}{h_{n-2}}(a_{n-1} - a_{n-2})$$

$$AC = B$$

$$\begin{array}{l} j=0 \\ j=1 \\ j=2 \\ j=3 \\ j=n \end{array} \begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & \dots & 0 \\ h_0 & 2(h_0+h_1) & h_1 & 0 & 0 & & 0 \\ 0 & h_1 & 2(h_1+h_2) & h_2 & 0 & & 0 \\ 0 & 0 & h_2 & 2(h_2+h_3) & h_3 & & 0 \\ 0 & 0 & 0 & 0 & 0 & \dots & 1 \end{array} \begin{array}{c} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_n \end{array} =$$

$$\begin{bmatrix} 0 \\ \frac{3}{h_1}(a_2 - a_1) & \frac{2}{h_0}(a_1 - a_0) \\ \frac{3}{h_2}(a_1 - a_2) & -\frac{2}{h_1}(a_2 - a_1) \\ \vdots \\ 0 \end{bmatrix}$$

This system involves only c_j 's as unknowns, since the value of h_j 's and a_j 's are given by the spacing of the nodes x_j and y -values at the nodes

$$\Rightarrow C = A^{-1} B$$

Note: Once the value of c_j 's are known, it is a simpler matter to find the remainder of the constant b_j 's and d_j 's

$$b_j = \frac{a_{j+1} - a_j}{h_j} - \frac{h_j}{3} (2c_j + c_{j+1}) \quad d_j = \frac{c_{j+1} - c_j}{3h_j}$$